

Aptitude, 2018 EC, entrance exam, version 2

Grade 12 · Aptitude · 2018 E.C.

Time allowed: 120 minutes

Questions: 57

Total marks: 57

Name _____ Grade _____ ID No. _____ Date _____

1. CALF : COW :: LAMB : ? (1)

A. Sheep	B. Horse
C. Deer	D. Goat
2. BOOK : PUBLISHER :: FILM : ? (1)

A. Writer	B. Editor
C. Director	D. Producer
3. Choose the word or phrase that is nearly the SAME IN MEANING as EXTROVERTED. (1)

A. Shy	B. Outgoing
C. Deceiver	D. Energetic
4. Choose the word or phrase that is nearly the SAME IN MEANING as ASSERTIVE. (1)

A. Passive	B. Aggressive
C. Confident	D. Hesitant
5. Choose the word or phrase that is nearly the SAME IN MEANING as DISINTERESTED. (1)

A. Impartial	B. Biased
C. Bored	D. Unconcerned
6. Choose the word or phrase that is nearly OPPOSITE IN MEANING as SUBJUGATE. (1)

A. Liberate	B. Oppress
C. Protest	D. Conquer
7. Choose the word or phrase that is nearly OPPOSITE IN MEANING as PENNYWISE. (1)

A. Selfish	B. Pleasant
C. Ungrateful	D. Extravagant
8. Choose the word or phrase that is nearly OPPOSITE IN MEANING as FRIENDLY. (1)

A. Hostile	B. Warm
C. Supportive	D. Affectionate
9. Find the ODD ONE OUT: Apple, Carrot, Grape, Banana (1)

A. Apple	B. Carrot
C. Grape	D. Banana
10. Find the ODD ONE OUT: London, Paris, New York, California (1)

A. London	B. Paris
C. New York	D. California

-
11. Find the ODD ONE OUT: Lion, Tiger, Elephant, Leopard (1)
- A. Lion B. Tiger
C. Elephant D. Leopard
-
12. The engineer's meticulous attention to detail in the design of the building ensured that everything was perfectly executed, from the architecture to the finishing touches. Choose a word that best substitutes 'meticulous'. (1)
- A. Reckless B. Thorough
C. Curious D. Imprecise
-
13. As soon as all the indispensable forms have been filled out, we will process our applications. Choose a word that best substitutes 'indispensable'. (1)
- A. Requisite B. Nonessential
C. Tentative D. Imprecise
-
14. Ali is all the time gluttonous to explore a wide range of topics, seeking to learn more and never satisfied with just the basics. Choose a word that best substitutes 'gluttonous'. (1)
- A. Voracious B. Excessive
C. Limited D. Disinterested
-
15. I've always been somewhat introverted, preferring to reflect quietly rather than engage in constant chatter with friends. Choose a word that best substitutes 'introverted'. (1)
- A. Outgoing B. Reserved
C. Indifferent D. Verbose
-
16. She wanted to impress all her dinner guests with the food she served, so she carefully studied the necessary culinary arts. Choose a word that best substitutes 'culinary'. (1)
- A. Architectural B. Literary
C. Cooking D. Gardening
-
17. Which of the following is grammatically correct for 'Neither the players nor the manager ___ ready for the next press conference'? (1)
- A. Neither the players nor the manager be ready for the next press conference.
B. Neither the players nor the manager were ready for the next press conference.
C. Neither the players nor the manager is ready for the next press conference.
D. Neither the players nor the manager have been ready for the next press conference.
-
18. Which of the following is grammatically correct for 'I personally believe the team ___ played ___ in the match'? (1)
- A. I personally believe the team has played wonderful in the match.
B. I personally believe the team have played wonderful in the match.
C. I personally believe the team has played wonderfully in the match.
D. I personally believe the team played wonderful in the match.
-
19. Which of the following is grammatically correct for 'Ali is one of the authorities who ___ responsible for the decision'? (1)
- A. Ali is one of the authorities who is responsible for the decision.
B. Ali is one of the authorities who are responsible for the decision.
C. Ali is one of the authority who is responsible for the decision.
D. Ali is one of the authority who are responsible for the decision.

-
20. If Keneni studies Chemistry on Saturday, which subject has she studied on Thursday? (1)
- | | |
|------------|------------------------------|
| A. Physics | B. English |
| C. Biology | D. Either Physics or English |
-
21. Which one of the following is NOT possible? (1)
- | | |
|--------------------------|------------------------|
| A. Chemistry on Thursday | B. Physics on Sunday |
| C. Physics on Thursday | D. Biology on Saturday |
-
22. Dida introduces Kaku as the daughter of the only son of his father's wife. How is Kaku related to Dida? (1)
- | | |
|-----------|-------------|
| A. Cousin | B. Niece |
| C. Aunt | D. Daughter |
-
23. After hearing about the new job opening in his field, Mark decided to apply immediately. Though it would mean moving to a new city, he was excited about the opportunity and felt it was a great career move. The job was also offering a higher salary than his current position, which made the decision easier for him. What is the most likely reason Mark decided to apply for the new job? (1)
- A. Mark enjoys living in his current city.
- B. Mark believes the new job is a better career opportunity.
- C. The new job requires him to relocate to a different country.
- D. Mark prefers jobs that offer lower salaries.
-
24. Meditation is a great way to enhance mental health and reduce stress, and you don't need a gym or special equipment to practice it. All you need is a quiet space, some time, and a comfortable place. Which of the following sentences best supports this idea? (1)
- A. Meditation is a simple, cost-effective practice that only needs a peaceful space to improve mental clarity.
- B. Gym workouts are worse for stress relief than meditation because it needs a professional instructor.
- C. Meditation requires special equipment like pillows or sound machines for effectiveness.
- D. You don't need a quiet space, some time and comfortable position for mediation.
-
25. *Mary Jonson was an obstinate and isolated child, who spent her early years in the lush landscapes of India. Following the death of her neglectful parents, she was sent to live with her reclusive uncle, Mr. Craven, in an enormous, deserted mansion in England. Initially, Mary was disinterested in her surroundings, sulking in her loneliness. However, a fateful discovery of a hidden door in the garden wall sparked her curiosity. As Mary offered into the long-forgotten garden, she encountered a wild, untamed world filled with neglected flora. Driven by a newfound purpose, she diligently nurtured the garden back to life, transforming it into a sanctuary of vibrant colors and charming scents. Through this process, she not only fostered the garden's revival but also cultivated friendships that helped her discover the joys of companionship, resilience, and personal growth.* How is Mary Jonson initially characterized in the passage? (1)
- | | |
|--------------------------|-------------------------------|
| A. Stubborn and solitary | B. Energetic and curious |
| C. Cheerful and sociable | D. Compassionate and friendly |
-

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- 26.** *Mary Jonson was an obstinate and isolated child, who spent her early years in the lush landscapes of India. Following the death of her neglectful parents, she was sent to live with her reclusive uncle, Mr. Craven, in an enormous, deserted mansion in England. Initially, Mary was disinterested in her surroundings, sulking in her loneliness. However, a fateful discovery of a hidden door in the garden wall sparked her curiosity. As Mary offered into the long-forgotten garden, she encountered a wild, untamed world filled with neglected flora. Driven by a newfound purpose, she diligently nurtured the garden back to life, transforming it into a sanctuary of vibrant colors and charming scents. Through this process, she not only fostered the garden's revival but also cultivated friendships that helped her discover the joys of companionship, resilience, and personal growth.* (1)
- What event led to Mary's relocation to her uncle's mansion?
- A. A family reunion
 - B. Her desire for adventure
 - C. The death of her parents
 - D. A travel to England for vacation
-

- 27.** *Mary Jonson was an obstinate and isolated child, who spent her early years in the lush landscapes of India. Following the death of her neglectful parents, she was sent to live with her reclusive uncle, Mr. Craven, in an enormous, deserted mansion in England. Initially, Mary was disinterested in her surroundings, sulking in her loneliness. However, a fateful discovery of a hidden door in the garden wall sparked her curiosity. As Mary offered into the long-forgotten garden, she encountered a wild, untamed world filled with neglected flora. Driven by a newfound purpose, she diligently nurtured the garden back to life, transforming it into a sanctuary of vibrant colors and charming scents. Through this process, she not only fostered the garden's revival but also cultivated friendships that helped her discover the joys of companionship, resilience, and personal growth.* (1)
- What does Mary find within the hidden garden?
- A. A magical fountain
 - B. An overgrown, neglected space
 - C. A place filled with animated animals
 - D. A structured and organized landscape
-

- 28.** *Mary Jonson was an obstinate and isolated child, who spent her early years in the lush landscapes of India. Following the death of her neglectful parents, she was sent to live with her reclusive uncle, Mr. Craven, in an enormous, deserted mansion in England. Initially, Mary was disinterested in her surroundings, sulking in her loneliness. However, a fateful discovery of a hidden door in the garden wall sparked her curiosity. As Mary offered into the long-forgotten garden, she encountered a wild, untamed world filled with neglected flora. Driven by a newfound purpose, she diligently nurtured the garden back to life, transforming it into a sanctuary of vibrant colors and charming scents. Through this process, she not only fostered the garden's revival but also cultivated friendships that helped her discover the joys of companionship, resilience, and personal growth.* (1)
- What transformation occurs in Mary as she cares for the garden?
- A. She gains a reputation as a gardener
 - B. She learns about gardening techniques
 - C. She becomes wealthier and more powerful
 - D. She discovers feelings of companionship and resilience

-
- 29.** *Every year, millions of monarch butterflies embark on an incredible migration from North America to central Mexico. This journey can span up to 3,000 miles, making it one of the longest migrations of any insect species. Monarchs travel in groups, known as clusters, and rely on specific environmental cues to guide them, such as temperature and the position of the sun. As they migrate, monarchs face numerous trials, including predators and changes in weather. They must find food sources, typically nectar from flowers, to sustain their journey. Interestingly, these butterflies are not born knowing the migration route; instead, it is believed they inherit the knowledge through instinct passed down from previous generations. Upon reaching their wintering grounds, the monarchs gather in vast numbers, covering the trees in a breathtaking display. This phenomenon not only underscores the resilience of the species but also highlights the importance of preserving their habitat. Protecting these migration pathways is crucial for the survival of monarchs, as environmental changes threaten their annual journey.* (1)

What distinguishes the migration of monarch butterflies from those of other insect species?

- A. It happens only in small clusters.
 - B. It relies solely on the position of the sun.
 - C. It occurs without the need for food sources.
 - D. It spans up to 3,000 miles, making it one of the longest migrations.
-

- 30.** *Every year, millions of monarch butterflies embark on an incredible migration from North America to central Mexico. This journey can span up to 3,000 miles, making it one of the longest migrations of any insect species. Monarchs travel in groups, known as clusters, and rely on specific environmental cues to guide them, such as temperature and the position of the sun. As they migrate, monarchs face numerous trials, including predators and changes in weather. They must find food sources, typically nectar from flowers, to sustain their journey. Interestingly, these butterflies are not born knowing the migration route; instead, it is believed they inherit the knowledge through instinct passed down from previous generations. Upon reaching their wintering grounds, the monarchs gather in vast numbers, covering the trees in a breathtaking display. This phenomenon not only underscores the resilience of the species but also highlights the importance of preserving their habitat. Protecting these migration pathways is crucial for the survival of monarchs, as environmental changes threaten their annual journey.* (1)

Which of the following best describes the challenges faced by monarch butterflies during migration?

- A. Lack of instinct on the migration route.
 - B. Sufficient food supply and safe climates.
 - C. Overpopulation at their wintering grounds.
 - D. Predators, adverse weather, and the search for nectar.
-

- 31.** *Every year, millions of monarch butterflies embark on an incredible migration from North America to central Mexico. This journey can span up to 3,000 miles, making it one of the longest migrations of any insect species. Monarchs travel in groups, known as clusters, and rely on specific environmental cues to guide them, such as temperature and the position of the sun. As they migrate, monarchs face numerous trials, including predators and changes in weather. They must find food sources, typically nectar from flowers, to sustain their journey. Interestingly, these butterflies are not born knowing the migration route; instead, it is believed they inherit the knowledge through instinct passed down from previous generations. Upon reaching their wintering grounds, the monarchs gather in vast numbers, covering the trees in a breathtaking display. This phenomenon not only underscores the resilience of the species but also highlights the importance of preserving their habitat. Protecting these migration pathways is crucial for the survival of monarchs, as environmental changes threaten their annual journey.* (1)

In what way is the knowledge of the migration route significant for the monarch butterflies?

- A. It allows them to adapt to new climates.
- B. It is vital for their survival during migration.
- C. It enables them to find better feeding grounds.
- D. It guarantees the success of their breeding cycle.

- 32.** Every year, millions of monarch butterflies embark on an incredible migration from North America to central Mexico. This journey can span up to 3,000 miles, making it one of the longest migrations of any insect species. Monarchs travel in groups, known as clusters, and rely on specific environmental cues to guide them, such as temperature and the position of the sun. As they migrate, monarchs face numerous trials, including predators and changes in weather. They must find food sources, typically nectar from flowers, to sustain their journey. Interestingly, these butterflies are not born knowing the migration route; instead, it is believed they inherit the knowledge through instinct passed down from previous generations. Upon reaching their wintering grounds, the monarchs gather in vast numbers, covering the trees in a breathtaking display. This phenomenon not only underscores the resilience of the species but also highlights the importance of preserving their habitat. Protecting these migration pathways is crucial for the survival of monarchs, as environmental changes threaten their annual journey.
- Based on the passage, what can be inferred about the role of environmental changes on the migration of monarch butterflies?
- A. Environmental changes have no significant impact on the migration of monarch butterflies.
B. Monarch butterflies are capable of adapting their migration routes quickly in response to environmental changes.
C. The survival of monarch butterflies during their migration is at risk due to environmental changes, making habitat preservation essential.
D. Monarch butterflies rely solely on instinct for their migration and do not require external environmental cues.
-
- 33.** The product of two numbers is 120 and the sum of their squares is 289. The sum of the numbers is: (1)
- A. 20
B. 23
C. 169
D. 529
-
- 34.** What is the value of X in the sequence 1, 5, 14, 30, 55, X, 140, 204? (1)
- A. 85
B. 86
C. 91
D. 75
-
- 35.** A library has an average of 510 visitors on Sundays and 240 on other days. The average number of visitors per day in a month of 30 days beginning with a Sunday is: (1)
- A. 250
B. 276
C. 280
D. 285
-
- 36.** Which of the following linear inequalities define the domain of the function $f(x, y) = \sqrt{4x - y + 6} + \ln(2x + y - 5)$? (1)
- A. $y \leq 4x + 6$ and $y > -2x + 5$
B. $y \leq 4x + 6$ and $y < -2x + 5$
C. $y \geq 4x + 6$ and $y \leq -2x + 5$
D. $y \geq 4x + 6$ and $y \geq -2x + 5$
-
- 37.** Find the odd one out from: 396, 427, 671, 264 (1)
- A. 396
B. 427
C. 671
D. 264
-
- 38.** The White, Red and Black balls in a packet are in the ratio of 5:3:2. How many Red and Black balls are there, if there are 40 White balls? (1)
- A. 30 reds and 50 black balls
B. 24 reds and 16 black balls
C. 16 reds and 24 black balls
D. 25 reds and 15 black balls
-
- 39.** If $2a + b = x$ and $-2a + b = y$, then which one of the following expressions is equal to ab ? (1)
- A. $(x + y)/2$
B. $(x - y)/4$
C. $(x^2 - y^2)/8$
D. $(x^2 + y^2)/8$

40. The average monthly income of P and Q is Birr 5050. The average monthly income of Q and R is 6250 and the average monthly income of P and R is 5200. The monthly income of P is: (1)

A. 3500 B. 4000
C. 4050 D. 5000

41. If an equilateral triangle and a square have the same perimeter, what is the ratio of the length of the side of the square to the length of the side of the equilateral triangle? (1)

A. 3:4 B. 1:2
C. 4:3 D. 2:1

42. Three numbers are in the ratio of 3:4:5 and the least common multiple is 2400. Their GCF is: (1)

A. 40 B. 80
C. 120 D. 200

43. A, B and C jointly thought of engaging themselves in a business venture. It was agreed that A would invest Birr 6500 for 6 months, B Birr 8400 for 5 months and C Birr 10,000 for 3 months. A wants to be the working member for which he was to receive 5% of the profit. The profit earned was Birr 7400. What is the share of B in the profit? (1)

A. Birr 1900 B. Birr 2660
C. Birr 2800 D. Birr 2840

44. In how many different ways can the letters of the word 'CORPORATION' be arranged so that vowels always come together? (1)

A. 810 B. 1440
C. 2880 D. 50,400

45. What is the probability of a number selected at random from the first 100 prime numbers being even? (1)

A. 99/100 B. 49/100
C. 1/100 D. 1/50

46. A committee consists of 6 men and 4 women. If two members are selected at random to form a subcommittee, what is the probability that both members are women? (1)

A. 1/15 B. 2/15
C. 3/10 D. 1/5

47. If the average age of husband, wife and their child 3 years ago was 27 years and that of wife and child 5 years ago was 20 years, the present age of the husband is: (1)

A. 35 years B. 40 years
C. 50 years D. 90 years

48. $(287)^2 + (269)^2 - 2(287 \times 269)$ is equal to: (1)

A. 534 B. 446
C. 354 D. 324

49. A piece of machinery is worth Birr 100,000 today and loses 25% of its value each year. How much will it be worth after 4 years? (1)

A. 75000 B. 41640.625
C. 31642 D. 31640.625

50. If x and y are integers such that $3x + 9 < 0$ and $-4y - 16 > 0$, then which one of the following is always true? (1)
- A. xy is positive
B. xy is an odd
C. xy is negative
D. xy is an even

51. For x is a real number, compare the following two quantities: Quantity A: $2x + 1$; Quantity B: $\sqrt{4x^2 + 4x + 1}$ (1)
- A. Quantity A is less than Quantity B
B. Quantity A is greater than Quantity B
C. Quantity A is equal to Quantity B
D. The relationship cannot be determined

52. Compare the following two quantities: Quantity A: The percent of income increase from 75 to 85 thousand birr; (1)
Quantity B: The percent of income increase from 200 to 220 thousand birr
- A. Quantity A is equal to Quantity B
B. Quantity A is greater than Quantity B
C. Quantity A is less than Quantity B
D. The relationship cannot be determined

53. The following table gives the sales of computers manufactured by a company over the years. Number of (1)
Different Types of Computers Sold by a Company over the Years (Numbers in Thousands)
- | Year | 4hp | 7hp | 32hp | 35hp | 55hp | Total |
|------|-----|-----|------|------|------|-------|
| 2000 | 75 | 144 | 114 | 102 | 108 | 543 |
| 2001 | 90 | 126 | 102 | 84 | 126 | 528 |
| 2002 | 96 | 114 | 75 | 105 | 135 | 525 |
| 2003 | 105 | 90 | 150 | 90 | 75 | 510 |
| 2004 | 90 | 75 | 135 | 75 | 90 | 465 |
| 2005 | 105 | 60 | 165 | 45 | 120 | 495 |
| 2006 | 115 | 85 | 160 | 100 | 145 | 605 |
- What was the approximate percentage increase in the sales of 55hp computers in 2006 compared to that in 2000?
- A. 28%
B. 31%
C. 33.26%
D. 34.26%

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- Which type of computer sales is maximum over seven years?
- A. 4hp
B. 7hp
C. 32hp
D. 35hp

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| 2005 | 105 | 60 | 165 | 45 | 120 | 495 |
| 2006 | 115 | 85 | 160 | 100 | 145 | 605 |
- What is the difference in the number of 35hp computers sold in 2001 and 2005?
- A. 24000
B. 28000
C. 35000
D. 39000

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| 2004 | 90 | 75 | 135 | 75 | 90 | 465 |
| 2005 | 105 | 60 | 165 | 45 | 120 | 495 |
| 2006 | 115 | 85 | 160 | 100 | 145 | 605 |
- Which type of computer had a continuous decrease in sales from 2000 to 2005?
- A. 4hp
B. 7hp
C. 32hp
D. 35hp

57. The following table gives the sales of computers manufactured by a company over the years. Number of
Different Types of Computers Sold by a Company over the Years (Numbers in Thousands)

Year	4hp	7hp	32hp	35hp	55hp	Total
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2001	90	126	102	84	126	528
2002	96	114	75	105	135	525
2003	105	90	150	90	75	510
2004	90	75	135	75	90	465
2005	105	60	165	45	120	495
2006	115	85	160	100	145	605

(1)

The percentage of 4hp computers sold to the total number of computers sold was maximum in the year?

- A. 2002
B. 2004
C. 2003
D. 2005

— END OF EXAMINATION —